

LLM Hallucination Detection via Uncertainty

SEDS 537 Machine Learning Term Project

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Abstract

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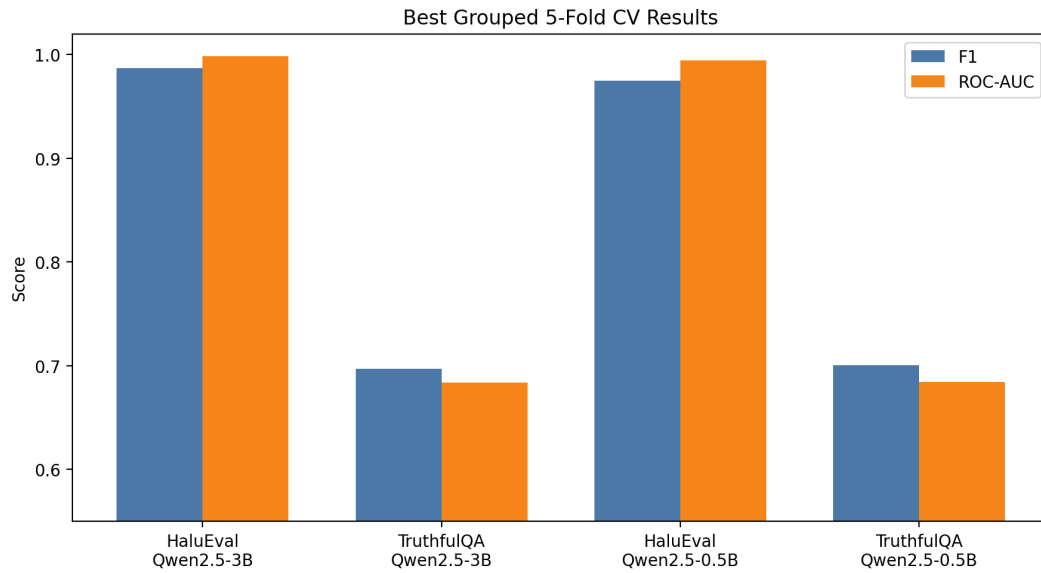


Figure 1: Grouped 5-fold cross-validation comparison between Qwen2.5-3B and Qwen2.5-0.5B feature extractors.

6.3 ROC Curves

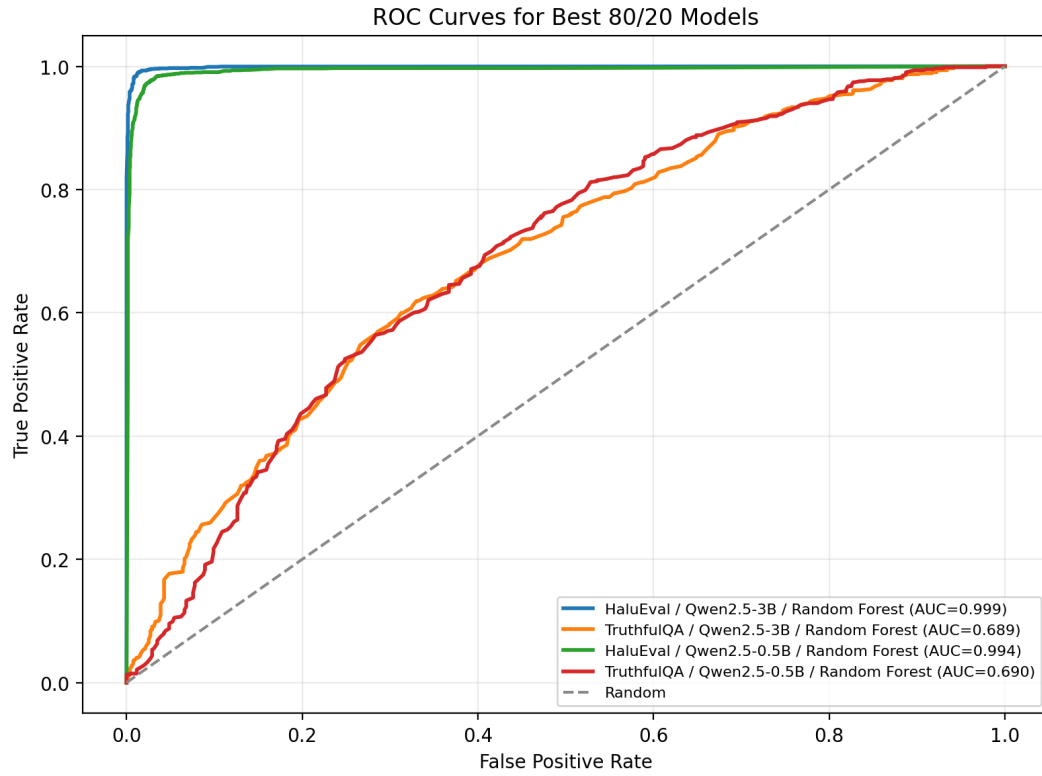


Figure 2: ROC curves for the best grouped 80/20 models.

7 Ablation Study

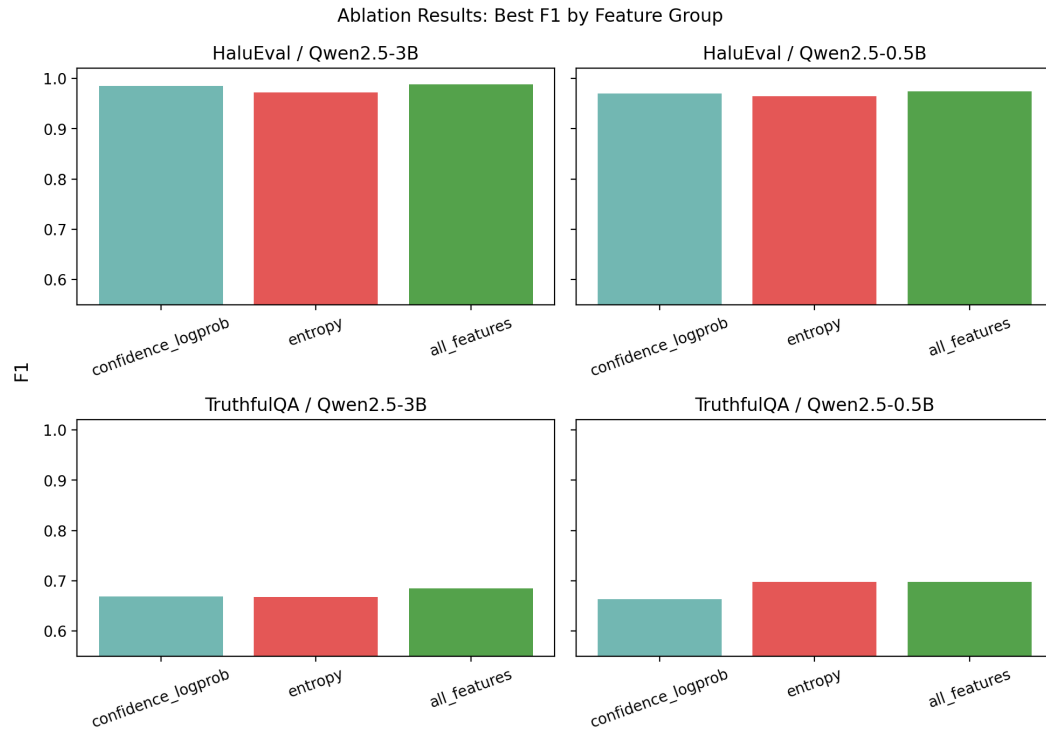


Figure 3: Ablation comparison across confidence/log-probability, entropy, and all feature groups.

8 Error Analysis

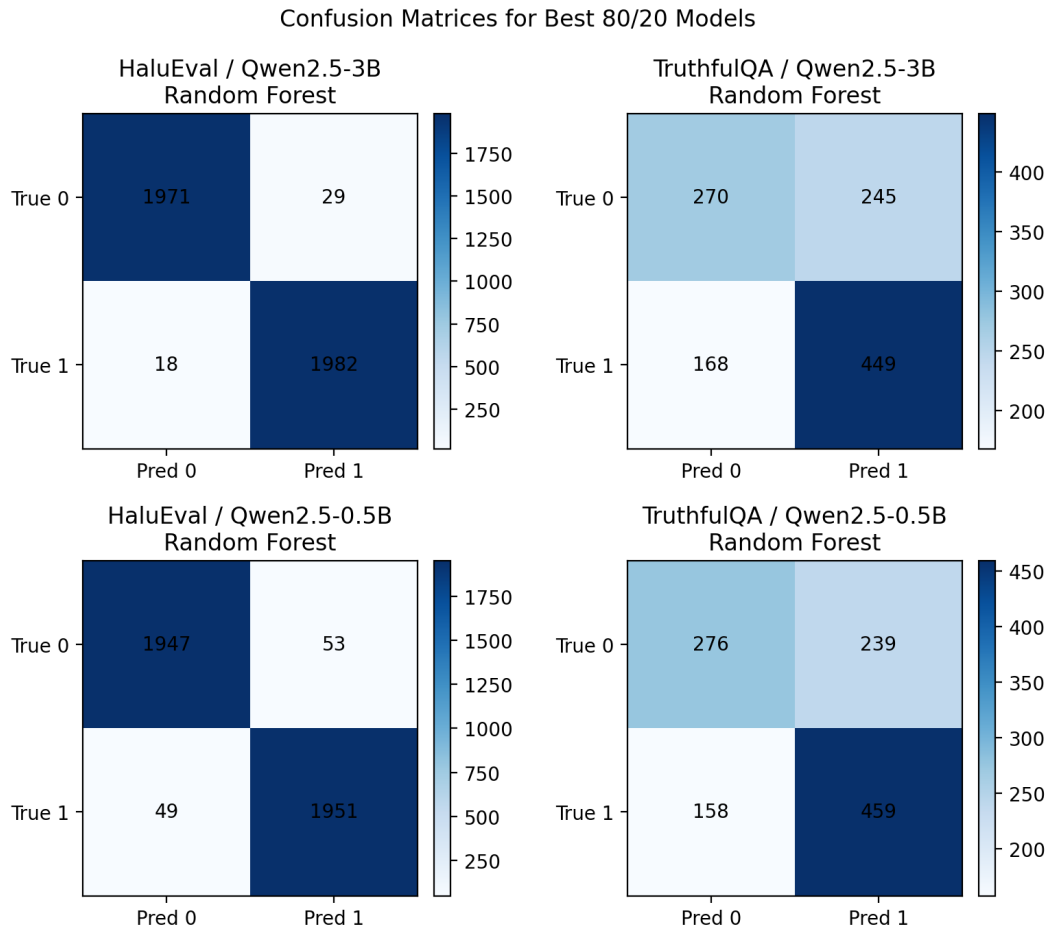


Figure 4: Confusion matrices for the best grouped 80/20 classifiers.

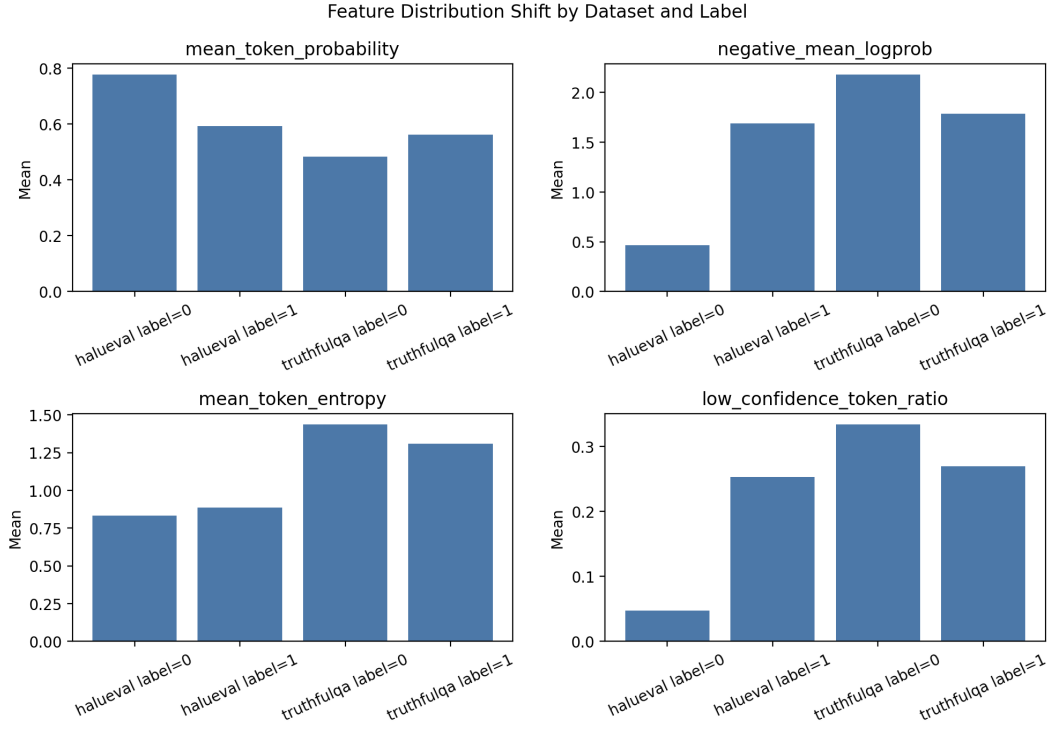


Figure 5: Selected uncertainty feature means by dataset and label.

9 Discussion and Limitations

10 Conclusion